

Md Samshad Rahman (MACS | M.Sc. Stats | B.Sc. CS)

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Professional summary:

Data Scientist with dual Master's degrees in Applied Statistics and Computer Science. I've spent the last few years building data pipelines, ML models, and turning raw, fragmented data into robust pipelines and actionable insights that drive real product decisions. I'm comfortable with the full process: from writing Python scripts, feature engineering and A/B test design to fixing broken reporting workflows and presenting findings to non-technical audiences. Actively use AI tools including Claude and LangChain to accelerate analysis and automate repetitive workflows.

Core competencies:

Data Quality & Governance:	Python (Pandas, NumPy, scikit-learn, spaCy), SQL (PostgreSQL, MySQL), Statistical Modelling, A/B Testing, Funnel Analysis, Customer Behavior Analysis, Feature Engineering, ETL Pipelines, Data Cleansing, Automated Validation.
ML & AI:	Machine Learning (LinearSVC, NLP, Classification), LangChain, LLM Fine-tuning, Computer Vision, Autonomous Agent Design.
Reporting & Visualization:	Power BI (DAX, data modelling), Tableau, Looker Studio, Dash/Plotly, Advanced Excel, JIRA, Confluence.
Cloud & Infrastructure:	AWS (EC2, S3, Lambda, Bedrock, Rekognition, EKS, and others), GCP (BigQuery, Cloud Functions, NLP API), Docker, Git, REST APIs, MS Office 365.

Relevant Projects & Experiences:

Dalhousie University, Halifax, Nova Scotia

AI-Powered Mindful Eating App (*Persuasive Computing Lab*)

[January 2025 – April 2025]

- Fine-tuned Llama 3.2 (3B) and deployed it on T4 GPUs, tuning memory and model parameters to hit the right balance of speed, accuracy, and cost.
- Built a safety layer to catch model hallucinations and validated outputs against 1,500 expert-reviewed inputs before they reached users.
- Integrated the Symanto NLP API to detect Big Five personality traits from user text, so the app gave advice that actually matched how each person thinks, not the same generic response for everyone.

Awarded Public Tenders Analysis Dashboard (*Visual Analytics*)

[September 2024 – December 2024]

- Fixed a broken NS government reporting process by writing Python scripts to ingest, clean, and standardize 10 years of raw Nova Scotia Open Data tender records.
- Wrote validation scripts that caught formatting errors and cleaned up ~35% of the noise.
- Built a custom intelligence tool using Dash/Plotly that clustered unstructured contract descriptions into coherent themes, cutting research time by ~40%.

Systems Analyst (*Data Integrity & Billing*), Munshi Engineers, Bangladesh

[February 2021 – April 2023]

- Managed billing infrastructure for two national power utilities (DPDC, BPDB), keeping high-volume financial data accurate and audit-ready.
- Engineered Python/SQL health checks to monitor consumption trends and flag statistical anomalies in billing data before invoice generation.
- Transformed raw operational logs into actionable monthly reports using Excel and SQL, enabling management to spot revenue leakage.

Python Developer, Smartbytes Ltd. Bangladesh

[February 2022 – September 2022]

Resume Matching with Job Descriptions (*German Job Market*)

- Automated resume screening by building an NLP engine that parsed unstructured resumes into structured PostgreSQL fields.
- Configured the API to auto-generate documentation (Swagger), reducing integration time for the frontend team.
- Benchmarked three models (Naive Bayes, Logistic Regression, LinearSVC) using A/B evaluation and F1 score analysis, improving match accuracy from ~65% to 92%.

Autonomous Agent API (*Personal Project*), github.com/samshad/autonomous-agent-api [February 2026 – March 2026]

- Built an autonomous agent pipeline using LangChain and a large language model backend, enabling multi-step reasoning and tool-use for structured task completion.
- Designed the API layer to expose agent capabilities as callable endpoints, demonstrating real-world LLM orchestration patterns.

Research Assistant, Cognitive & Behavioral Data Science Lab, UI University, Bangladesh [October 2021 – March 2022]

Eating Preference Analysis Based on Human Personality Traits – Big5 (Data Science)

- Built an NLP pipeline with spaCy and IBM Watson to extract Big Five personality traits, cutting dataset noise by ~45% and improving model accuracy by ~30%.
- Wrote sanitization scripts to strip noise like emojis and broken URLs from thousands of raw posts and shape them into a clean, analysis-ready dataset.
- Applied Big Five trait segmentation model to support targeted content recommendation research that is directly applicable to personalized user experience design.

Software Engineer (Revenue Operations), Free Pixel Games Limited, Bangladesh

[November 2020 – May 2021]

Unified Ad-Service Revenue Dashboard

- Designed a unified dimensional data model consolidating revenue data from 5+ ad networks into a single MySQL warehouse, enabling consistent KPI reporting across all monetization channels.
- Automated the daily revenue reconciliation workflow, reducing manual data consolidation time by >80% (from 2 hours to 15 mins).
- Built real-time alerts that fired whenever eCPM dropped more than 10%, catching revenue issues before they snowballed.

Other Experience:

Microsoft Senior Advisor, Best Buy Canada Ltd. Halifax, NS

[October 2023 – Present]

- Put together a repeatable training approach for Microsoft 365 products so new team members got up to speed faster and gave customers consistent advice.
- Tracked individual and team KPIs using weekly sales data to identify performance trends, flag gaps early, and share findings with team members in a way that actually changed how they approached their next shift.
- Consistently outperformed the store's revenue-per-hour benchmark by ~35% by actually listening to what customers needed before recommending anything.
- Contributed ~29% of total departmental revenue by matching customers to the right products based on their actual technical needs.

Education:

- **Master of Applied Computer Science** [September 2023 – May 2025]
Dalhousie University, Halifax, NS
- **Master's in Applied Statistics and Data Science** [November 2019 – April 2022]
Jahangirnagar University, Bangladesh
- **BSc. in Computer Science and Software Engineering** [January 2015 – January 2020]
American International University-Bangladesh